Python Placement Readiness

An interactive exercise book with 100 challenges to sharpen your skills.

All Questions | Level 1: Beginner | Level 2: Intermediate | Level 3: Advanced

Level 1

Q1: Divisibility Filter

Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included). The numbers obtained should be printed in a comma-separated sequence on a single line.

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Level 1

Q2: Factorial Calculation

Write a program which can compute the factorial of a given number. The result should be printed on a single line.

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Level 1

Q3: Dictionary of Squares

With a given integral number n, write a program to generate a dictionary that contains (i, i*i) such that i is an integral number between 1 and n (both included). Then the program should print the dictionary.

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Level 1

Q4: List and Tuple Generator

Write a program which accepts a sequence of comma-separated numbers from the console and generates a list and a tuple which contains every number.

Show Details ▼

Level 1

Q5: Simple Class Definition

Define a class which has at least two methods:
1. getString: to get a string from console input.
2. printString: to print the string in upper case.

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Level 1

Q6: Method to Calculate Square

Write a method which can calculate the square value of a number.

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Level 1

Q7: Documenting Functions

Python has a built-in document function `__doc__` for every function. Write a program to print the documents for built-in functions like `abs()` and `int()`, and also add and print a document string for your own simple function.

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Level 1

Q8: Class and Instance Parameters

Define a class, which has a class parameter and a same-named instance parameter. Show the difference when accessing the class parameter versus the instance parameter.

Show Details ▼

Level 1

Q56: Basic Function Sum

Define a function which can compute the sum of two numbers and return the result. Print the result of calling the function.

Show Details ▼

Level 1

Q57: Integer to String Conversion

Define a function that can convert an integer into a string and print it in the console.

Show Details ▼

Level 1

Q58: String Concatenation

Define a function that can accept two strings as input and concatenate them, then print the result.

Show Details ▼

Level 1

Q59: Length Comparison and Printing

Define a function that can accept two strings as input and print the string with the maximum length. If two strings have the same length, the function should print both strings line by line.

Show Details ▼

Level 1

Q60: Even or Odd Checker

Define a function that can accept an integer number as input and print the message "It is an even number" if the number is even, otherwise print "It is an odd number".

Show Details ▼

Level 1

Q61: Dictionary of Squares (1 to 3)

Define a function which can print a dictionary where the keys are numbers between 1 and 3 (both included) and the values are the square of the keys.

Show Details ▼

Level 1

Q62: Print List (1 to 20, Squares)

Define a function which can generate and print a list where the values are the square of numbers between 1 and 20 (both included).

Show Details ▼

Level 1

Q63: Print First 5 Elements

Define a function which can generate a list where the values are the square of numbers between 1 and 20 (both included). Then the function needs to print the first 5 elements in the list.

Show Details ▼

Level 1

Q64: Print Last 5 Elements

Define a function which can generate a list where the values are the square of numbers between 1 and 20 (both included). Then the function needs to print the last 5 elements in the list.

Show Details ▼

Level 1

Q65: Print All Except First 5

Define a function which can generate a list where the values are the square of numbers between 1 and 20 (both included). Then the function needs to print all values except the first 5 elements in the list.

Show Details ▼

Level 1

Q66: Generate and Print Tuple

Define a function which can generate and print a tuple where the values are the square of numbers between 1 and 20 (both included).

Show Details ▼

Level 1

Q67: Tuple Halves

With a given tuple (1,2,3,4,5,6,7,8,9,10), write a program to print the first half of the values in one line and the last half of the values in a separate line.

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Level 1

Q68: Even Numbers in a Tuple

Write a program to generate and print another tuple whose values are the even numbers in the given tuple (1,2,3,4,5,6,7,8,9,10).

Show Details ▼

Level 1

Q69: Case-Insensitive Yes/No Check

Write a program which accepts a string as input to print "Yes" if the string is "yes", "YES", or "Yes", otherwise print "No".

Show Details ▼

Level 1

Q70: Sum of String Integers

Define a function that can receive two integral numbers in string form and compute their sum, then print it in the console.

Show Details ▼

Level 2

Q9: Formula Calculation and Rounding

Write a program that calculates and prints the value according to the given formula: Q = sqrt((2 * C * D)/H). Following are the fixed values: C = 50, H = 30. D is the variable whose values should be input to your program in a comma-separated sequence. Round the output off to its nearest integer.

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Level 2

Q10: 2D Array Generation

Write a program which takes 2 digits, X,Y as input and generates a 2-dimensional array. The element value in the i-th row and j-th column of the array should be i * j.

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Level 2

Q11: Alphabetical Word Sorting

Write a program that accepts a comma separated sequence of words as input and prints the words in a comma-separated sequence after sorting them alphabetically.

Show Details ▼

Level 2

Q12: Capitalizing Input Lines

Write a program that accepts a sequence of lines as input (until an empty line is entered) and prints the lines after making all characters in the sentence capitalized.

Show Details ▼

Level 2

Q13: Removing Duplicates and Sorting Words

Write a program that accepts a sequence of whitespace-separated words as input and prints the words after removing all duplicate words and sorting them alphanumerically.

Show Details ▼

Level 2

Q14: Binary Divisibility Check

Write a program which accepts a sequence of comma-separated 4-digit binary numbers as its input and then checks whether they are divisible by 5 or not. The numbers that are divisible by 5 are to be printed in a comma-separated sequence.

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Level 2

Q15: All Even Digits Filter

Write a program which will find all such numbers between 1000 and 3000 (both included) such that each digit of the number is an even number. The numbers obtained should be printed in a comma-separated sequence on a single line.

Show Details ▼

Level 2

Q16: Counting Letters and Digits

Write a program that accepts a sentence and calculates the number of letters and digits.

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Level 2

Q17: Counting Upper and Lower Case Letters

Write a program that accepts a sentence and calculates the number of upper case letters and lower case letters.

Show Details ▼

Level 2

Q18: Sum of Repeated Digits

Write a program that computes the value of a + aa + aaa + aaaa with a given digit a as input.
Show Details ▼
Level 2

Q19: List Comprehension for Odd Squares

Use a list comprehension to square each odd number in a list. The list is input by a sequence of comma-separated numbers.

Show Details ▼
Level 2

Q20: Bank Account Transaction Log

Write a program that computes the net amount of a bank account based on a transaction log from console input. The transaction log format is `D ` (Deposit) or `W ` (Withdrawal). Stop reading input when an empty line is supplied.

Show Details ▼
Level 2

Q21: Class for Nationality

Define a class named `American` which has a static method called `printNationality`.

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Level 2

Q22: Inheritance (American and NewYorker)

Define a class named `American` and its subclass `NewYorker`. Print the objects of both classes to demonstrate the inheritance relationship.

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Level 2

Q23: Class for Circle Area

Define a class named `Circle` which can be constructed by a radius. The `Circle` class has a method which can compute the area (π * r^2). Use π ≈ 3.14.

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Level 2

Q24: Class for Rectangle Area

Define a class named `Rectangle` which can be constructed by a length and width. The `Rectangle` class has a method which can compute the area.

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Level 2

Q25: Polymorphism with Area

Define a class named `Shape` and its subclass `Square`. The `Square` class has an `init` function that takes a length as an argument. Both classes must have an `area` method, where `Shape`'s area is 0 by default.

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Level 2

Q71: Filter Even Numbers (filter function)

Write a program which can filter even numbers in a list by using the `filter()` function. The list is: `[1,2,3,4,5,6,7,8,9,10]`.

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Level 2

Q72: Square Elements (map function)

Write a program which can use `map()` to make a list whose elements are the square of elements in `[1,2,3,4,5,6,7,8,9,10]`.

Show Details ▼

Level 2

Q73: Map and Filter: Even Squares

Write a program which can use `map()` and `filter()` to make a list whose elements are the square of even numbers in `[1,2,3,4,5,6,7,8,9,10]`.

Show Details ▼

Level 2

Q74: Filter Even Numbers in Range

Write a program which can use `filter()` to make a list whose elements are even numbers between 1 and 20 (both included).

Show Details ▼

Level 2

Q75: Map Squares in Range

Write a program which can use `map()` to make a list whose elements are the square of numbers between 1 and 20 (both included).

Show Details ▼

Level 3

Q26: Password Validation with Regex

A website requires users to input a password. Write a program to check the validity of passwords based on the following criteria: At least 1 [a-z], 1 [0-9], 1 [A-Z], 1 [$#@]. Minimum length: 6, Maximum length: 12. Your program should accept a sequence of comma-separated passwords and print only the valid ones.

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Level 3

Q27: Multi-Key Tuple Sorting

You are required to write a program to sort a list of (name, age, height) tuples by ascending order. The sort criteria priority is: Name -> Age -> Height. The tuples are input by console, one per line (until an empty line).

Show Details ▼

Level 3

Q28: Generator for Divisibility by 7

Define a class with a generator method which can iterate the numbers, which are divisible by 7, between a given range 0 and n.

Show Details ▼

Level 3

Q29: Robot Movement Distance

A robot moves in a plane starting from (0,0). The robot can move toward UP, DOWN, LEFT, and RIGHT. Compute the distance from the current position to the origin. If the distance is a float, print the nearest integer. Input lines stop when an empty line is entered.

Show Details ▼

Level 3

Q30: Word Frequency Counter

Write a program to compute the frequency of words from the input. The output should be sorted alphabetically by word and formatted as word:frequency.

Show Details ▼

Level 3

Q31: Extracting Username from Email

Assuming we have email addresses in the "username@companyname.com" format, write a program to print the user name of a given email address.

Show Details ▼

Level 3

Q32: Extracting Company Name from Email

Assuming we have email addresses in the "username@companyname.com" format, write a program to print the company name of a given email address.

Show Details ▼

Level 3

Q33: Filtering Digits with Regex

Write a program which accepts a sequence of words separated by whitespace as input to print the words composed of digits only.

Show Details ▼

Level 3

Q34: Sum of Series i/(i+1)

Write a program to compute 1/2 + 2/3 + 3/4 + ... + n/(n+1) with a given integer n input by console (n>0). Print the result with two decimal places.

Show Details ▼

Level 3

Q35: Recursive Function f(n)

Write a program to compute the value of f(n) defined by: f(n) = f(n-1) + 100 when n > 0 and f(0) = 0, with a given integer n input by console (n>0).

Show Details ▼

Level 3

Q36: Fibonacci Sequence f(n)

The Fibonacci Sequence is computed based on the formula: f(n)=0 if n=0, f(n)=1 if n=1, f(n)=f(n-1)+f(n-2) if n>1. Please write a program to compute the value of f(n) with a given integer n input by console.

Show Details ▼

Level 3

Q37: Fibonacci Sequence Generation

Use list comprehension and a recursive function to print the Fibonacci Sequence in comma-separated form for all values from f(0) up to f(n) where n is input by console.

Show Details ▼

Level 3

Q38: Even Number Generator

Please write a program using a generator to print the even numbers between 0 and n (both inclusive) in comma-separated form, where n is input by console.

Show Details ▼

Level 3

Q39: Divisible by 5 and 7 Generator

Please write a program using a generator to print the numbers which can be divisible by 5 and 7 between 0 and n (both inclusive) in comma-separated form, where n is input by console.

Show Details ▼

Level 3

Q40: Simple Expression Evaluation

Please write a program which accepts a basic mathematical expression (e.g., 35+3, 5*6) from the console and prints the evaluation result.

Show Details ▼

Level 3

Q41: Binary Search Implementation

Please write a binary search function which searches for an item in a sorted list. The function should return the index of the element to be searched in the list, or -1 if not found.

Show Details ▼

Level 3

Q42: Sentence Generation from Lists

Please write a program to generate all sentences where the subject is in ["I", "You"], the verb is in ["Play", "Love"], and the object is in ["Hockey","Football"].

Show Details ▼

Level 3

Q43: List Comprehension Filter

Please write a program to print the list after removing all even numbers in [5,6,77,45,22,12,24] using list comprehension.

Show Details ▼

Level 3

Q44: List Comprehension: Multi-Divisibility Removal

By using list comprehension, please write a program to print the list after removing numbers which are divisible by both 5 and 7 in [12,24,35,70,88,120,155].

Show Details ▼

Level 3

Q45: List Comprehension: Index Removal

By using list comprehension, please write a program to print the list after removing the 0th, 2nd, 4th, and 6th numbers in [12,24,35,70,88,120,155].

Show Details ▼

Level 3

Q46: 3D Array Generation

By using list comprehension, please write a program to generate a 3x5x8 3D array whose each element is 0.

Show Details ▼

Level 3

Q47: Set Intersection

With two given lists [1,3,6,78,35,55] and [12,24,35,24,88,120,155], write a program to make a new list whose elements are the intersection of the two given lists.

Show Details ▼

Level 3

Q48: Remove Duplicates (Order Preserved)

With a given list [12,24,35,24,88,120,155,88,120,155], write a program to print this list after removing all duplicate values, with the original order preserved.

Show Details ▼

Level 3

Q49: Character Frequency Counter

Please write a program which counts and prints the numbers of each character in a string input by the console. The output should be formatted as character,count, one pair per line.

Show Details ▼

Level 3

Q50: Chinese Puzzle Solver

Write a program to solve a classic ancient Chinese puzzle: We count 35 heads and 94 legs among the chickens and rabbits in a farm. How many rabbits and how many chickens do we have?

Show Details ▼

Level 3

Q51: String Reversal (Slicing)

Please write a program which accepts a string from the console and prints it in reverse order.

Show Details ▼

Level 3

Q52: Even Index Characters (Slicing)

Please write a program which accepts a string from the console and prints the characters that have even indexes (index 0, 2, 4, ...).

Show Details ▼

Level 3

Q53: List Permutations

Please write a program which prints all possible permutations of the list [1,2,3].

Show Details ▼

Level 3

Q54: Handling ZeroDivisionError

Write a function to compute 5/0 and use try/except to catch the exceptions and print a friendly message.

Show Details ▼

Level 3

Q55: Custom Exception Class

Define a custom exception class named `MyError` which takes a string message as an attribute and inherits from the base `Exception` class.

Show Details ▼

Level 3

Q76: Timeit for Code Execution

Please write a program to print the running time of execution of the command "1+1" repeated 100 times using the `timeit` module.

Show Details ▼

Level 3

Q77: List Shuffle

Please write a program to shuffle and print the list [3,6,7,8] using the `random` module.

Show Details ▼

Level 3

Q78: Random Float (10 to 100)

Please generate a random float where the value is between 10 and 100 (inclusive) using the `random` module.

Show Details ▼

Level 3

Q79: Random Even Number (0 to 10)

Please write a program to output a random even number between 0 and 10 (inclusive) using the `random` module and list comprehension.

Show Details ▼

Level 3

Q80: Random Divisible by 5 and 7

Please write a program to output a random number, which is divisible by 5 and 7, between 0 and 200 (inclusive) using the `random` module and list comprehension.

Show Details ▼

Level 3

Q81: Random Sample (100 to 200)

Please write a program to generate a list with 5 random numbers between 100 and 200 (inclusive) using `random.sample()`.

Show Details ▼

Level 3

Q82: Random Even Sample

Please write a program to randomly generate a list with 5 even numbers between 100 and 200 (inclusive).

Show Details ▼

Level 3

Q83: Random Divisible Sample

Please write a program to randomly generate a list with 5 numbers, which are divisible by 5 and 7, between 1 and 1000 (inclusive).

Show Details ▼

Level 3

Q84: Random Integer in Range

Please write a program to randomly print an integer number between 7 and 15 (inclusive) using `random.randrange()`.

Show Details ▼

Level 3

Q85: List Comprehension: Index Removal (Specific)

By using list comprehension, please write a program to print the list after removing the 0th, 4th, and 5th numbers in [12,24,35,70,88,120,155].

Show Details ▼

Level 3

Q86: List Comprehension: Value Removal (Duplicate)

By using list comprehension, please write a program to print the list after removing all instances of the value 24 in [12,24,35,24,88,120,155].

Show Details ▼

Level 3

Q87: Class Inheritance (Male/Female)

Define a class `Person` and its two child classes: `Male` and `Female`. All classes have a method "getGender" which can return "Male" for Male class, "Female" for Female class, and "Unknown" for Person.

Show Details ▼

Level 3

Q88: Unicode String Definition

Print a unicode string "hello world".

Show Details ▼

Level 3

Q89: ASCII to Unicode Conversion (utf-8)

Write a program to read an ASCII string from console input and convert it to a unicode string encoded by utf-8, then print the unicode string.

Show Details ▼

Level 3

Q90: Python Source Encoding Comment

Write a special comment that is commonly used to indicate a Python source code file is in unicode (specifically, utf-8 encoding). Print the comment as a string.

Show Details ▼

Level 3

Q91: Data Compression/Decompression (zlib)

Please write a program to compress and decompress the string "hello world!hello world!hello world!hello world!" using the zlib module. Print both the compressed and decompressed results.

Show Details ▼

Level 3

Q92: Raising a RuntimeError

Please write a program that will explicitly raise a RuntimeError exception with the message "Something went wrong".

Show Details ▼

Level 3

Q93: Fibonacci Function (Iterative)

The Fibonacci Sequence is defined as Fn = Fn-1 + Fn-2. Write a function to compute Fn using an iterative approach to avoid performance issues for large n. Compute F(10).

Show Details ▼

Level 3

Q94: Dictionary Keys Only

Define a function which can generate a dictionary where the keys are numbers between 1 and 20 (both included) and the values are the square of keys. The function should then print the keys only, one per line.

Show Details ▼

Level 3

Q95: Dictionary Values Only

Define a function which can generate a dictionary where the keys are numbers between 1 and 20 (both included) and the values are the square of keys. The function should then print the values only, one per line.

Show Details ▼

Level 3

Q96: Assert for Evenness

Please write `assert` statements to verify that every number in the list [2,4,6,8] is even. Your program should run without output if successful.

Show Details ▼

Level 3

Q97: Assert for Failure

Write a program that uses an `assert` statement to check if a user-input number is greater than 10. If the input is less than or equal to 10, the program should terminate with an AssertionError.

Show Details ▼

Level 3

Q98: String Formatting (f-strings)

Given the variables name = "John" and age = 30, use an f-string to print the sentence: "Hello, my name is John and I am 30 years old."

Show Details ▼

Level 3

Q99: List Comprehension: Flattening a List

Given a nested list [[1, 2], [3, 4], [5, 6]], use list comprehension to create a flattened list [1, 2, 3, 4, 5, 6].

Show Details ▼

Level 3

Q100: Simple Lambda Function

Define a simple lambda function that takes two arguments, x and y, and returns their product. Use this lambda function to calculate the product of 12 and 5.

Show Details ▼

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